

Real-to-Sim Scene Generation from a Single Photograph for Robot Simulation

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Abstract— We propose a practical Real-to-sim pipeline for generating simulation-ready initial scenes from a single photograph. Our method focuses on movable, interactive objects and integrates the following processes: object extraction; image-conditioned 3D asset Generation; coarse semantic and physical attribute initialization; orientation correction; and automatic layout Generation. Instead of targeting full scene reconstruction, our goal is to provide reusable 3D assets and a plausible initial layout for robot simulations. Preliminary feasibility is demonstrated by experimental results: the mean normalized size error was 11.96%, and a basic table-cup placement task preserved the target support relation in all 50 trials with zero observed collisions. Our framework can reduce the manual effort needed to prepare simulation-ready assets and layouts from real observations.

Keywords—Real-to-Sim, digital twin, single-image asset construction, simulation-ready scene initialization, physically plausible attributes, automatic layout

I. INTRODUCTION

In recent years, simulation-centered development has become an important methodology for embodied AI and intelligent robotics. Instead of training and validating robot behaviors only in the physical world, researchers increasingly rely on digital-twin environments to prototype perception, navigation, manipulation, and task planning under safe, repeatable, and cost-effective conditions [1]–[4]. This trend is particularly important in workflows that combine generative AI, vision-language models, large language models, and reinforcement learning, where large numbers of trials are often required before deployment in real environments [2], [5]. Therefore, rapidly building simulation-ready environments from real-world observations has become an important technical requirement for practical robot development [1], [3].

However, constructing simulation environments that are both realistic enough for robot testing and efficient enough for repeated use remains difficult. A practical robot-simulation scene requires not only visual plausibility, but also reusable 3D assets, coarse geometric consistency, support relations, and basic physical attributes for collision checking, navigation, and interaction. Existing approaches address only part of this requirement. Manual scene modeling is labor-intensive and does not scale well. Neural scene reconstruction methods such as NeRF and Gaussian-splatting-based approaches can produce visually convincing scene representations [6], [7], but they are not primarily designed for reusable object-centric asset construction or simulation-oriented scene editing. Recent image-to-3D and text-to-3D methods improve object modeling efficiency [8]–[11], yet they often provide limited control over object orientation, metric scale, physical attributes, and structured placement constraints required for robot simulation.

Within a digital-twin pipeline, this problem should be addressed from three coupled aspects: **space**, **objects**, and

scene layout. First, the static structure of the environment should be represented in a form that can be easily imported into simulators. Second, movable objects should be transformed into reusable 3D assets. Third, these assets should be placed according to semantically meaningful and physically plausible relations so that the resulting scene can serve as an initial state for robot training or verification. In our project, this requirement arises directly from the need to construct virtual environments for downstream robot perception, obstacle handling, and navigation experiments.

To address this need, this paper presents a practical Real-to-Sim scene-Generation pipeline that converts a real-world observation into a simulation-ready initial scene for robot simulation. Rather than pursuing full high-fidelity reconstruction of an entire environment, the proposed method emphasizes rapid Generation of reusable scene components and plausible initial spatial arrangement. The pipeline integrates movable-object extraction, image completion, image-conditioned 3D asset Generation, coarse semantic and physical attribute assignment, orientation correction, and retrieval-based automatic placement. The goal of the proposed method is not photorealistic reconstruction itself, but practical simulation initialization for digital-twin-based robotics.

The main difference between our method and existing methods is that conventional scene construction relies heavily on manual measurement and modeling, while our method uses a large language model to estimate object dimensions and coarse physical attributes automatically, reducing time and labor costs. The contributions of this paper are summarized as follows:

1. **We propose a practical Real-to-Sim pipeline for simulation-oriented scene initialization, which converts real observations into reusable 3D assets and an initial scene layout for robot simulation.**
2. **We propose a modular framework that integrates object extraction, image-conditioned 3D Generation, coarse semantic/physical attribute assignment, orientation correction, and retrieval-based placement.**
3. **We provide an initial quantitative evaluation of the proposed framework and discuss its current scope and limitations for digital-twin-based robot simulation.**

The remainder of this paper is organized as follows. Section II describes the system architecture, including asset construction, orientation correction, and automatic layout Generation. Section III presents the experimental setup and quantitative results. Section IV concludes the paper and discusses future directions.

II. SYSTEM ARCHITECTURE

Our framework converts a real-world observation into a simulation-ready initial scene for robot simulation. Rather than reconstructing a complete scene with high geometric fidelity, it focuses on generating reusable interactive object

assets and arranging them into a plausible initial layout. As illustrated in **Fig. 1**, the overall pipeline starts from a single RGB image and consists of two stages: **interactive asset construction** and **automatic scene layout Generation**. The final output is a reusable asset set together with a simulation-ready initial layout.

A. Interactive Asset Construction from a Single Photo

Given an input image, the system first identifies movable objects relevant to later interaction in simulation [13], [14], while excluding static background regions. The extracted object regions are removed from the original image, and an image-completion module restores the missing background areas [15]. Each cropped object image is then converted into a textured 3D asset by an image-conditioned 3D Generation model [10], as shown in **Fig. 1**.

To support later simulator initialization, the generated assets are assigned coarse semantic and physical metadata, including object category, approximate size, and simplified physical properties. These attributes are used as simulation-oriented initialization rather than exact physical measurements. In addition, predefined rule-based priors, such as plausible size ranges and category-dependent defaults, are used to improve robustness. The resulting assets are stored in an interactive asset library and later reused in the layout stage. This design emphasizes practicality and modularity: even if some estimates are imperfect, the framework still provides a usable initial approximation for simulation.

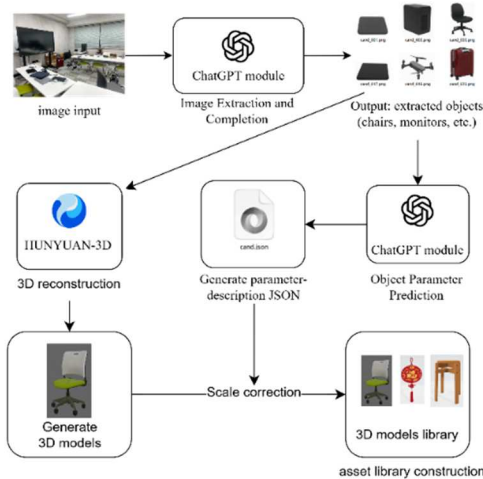


Fig. 1. Proposed pipeline for interactive 3D asset library construction from a single image.

B. Model Orientation Correction

A practical difficulty of current image-to-3D Generation is that the generated assets do not always share a consistent semantic orientation [8],[10]. Even when the geometry looks reasonable, the front direction or upright direction may be incorrect, which can degrade downstream placement quality. This issue is particularly problematic for asymmetric daily objects and furniture whose orientation affects how they should appear in a scene.

To reduce this problem, the framework applies a lightweight VLM-based orientation-correction procedure, as illustrated in **Fig. 2**. For each generated asset, four rendered views are produced at 90-degree intervals around the vertical axis. These candidate views are then evaluated by a vision-language model [17], which predicts the most plausible front-

facing direction. The corresponding rotation is stored as the canonical heading of the asset and used in later placement. Although this discrete four-view strategy is not a complete pose-estimation method, it serves as a practical correction mechanism that improves downstream layout consistency.

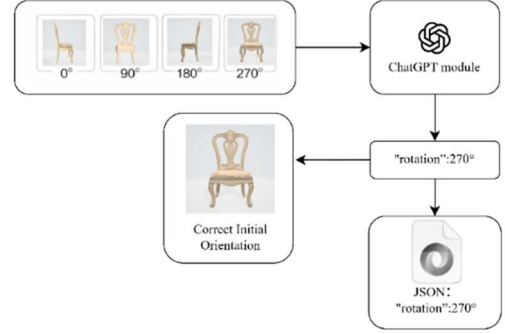


Fig. 2. VLM-based orientation correction for generated 3D assets

C. Automatic Scene Layout Generation

After the asset library is constructed, the system generates an initial scene layout from either a simple text instruction or a layout input. As shown in **Fig. 3**, the system first builds a structured layout JSON containing object categories, coarse relative positions, and support or parent-child relations. Large-language-model analysis [18] is used to supplement semantic attributes and relation descriptions. These textual descriptions are then encoded into semantic embeddings [16] and matched against the asset library to retrieve reusable 3D assets with compatible semantics [12], [16].

During layout Generation, the retrieved assets are aligned according to their support structure and approximate bounding-box dimensions, while simple collision checks are applied. For example, when one object is described as being placed on another, the system aligns the lower surface of the upper object with the supporting surface of the lower object while maintaining a small clearance. The purpose of this stage is not full scene optimization, but efficient Generation of a plausible and collision-aware initial arrangement for robot simulation.

The final output of the proposed framework consists of 1) reusable 3D object assets with coarse semantic and physical attributes, and 2) an initial scene layout under simple relational and geometric constraints. This output can be directly introduced into robot simulators such as the Isaac Sim or Genesis, as an initial state for perception, navigation, and interaction experiments, which is consistent with the broader digital-twin workflow of this project [1], [3], [4].

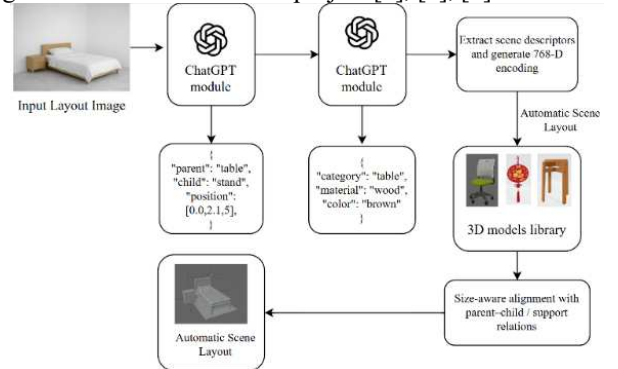


Fig. 3. Automatic scene layout using LLM-generated constraints and asset library retrieval

III. EXPERIMENTAL EVALUATION

A. Evaluation of Asset Generation and Scale Error

Using a real household scene photograph as input, as shown in **Fig. 4(a)**, we evaluated whether the proposed pipeline can generate object-level assets that are sufficiently accurate for simulation initialization. **Fig. 4(b)** shows the extracted movable objects, and **Fig. 4(c)** presents the generated 3D assets arranged for later simulation use. The purpose of this experiment is not to assess exact geometric reconstruction, but to examine whether the pipeline can provide stable object extraction, usable 3D assets, and reasonable scale initialization.

Based on records from 86 instances in 9 categories, the predicted dimensions $(\hat{x}, \hat{y}, \hat{z})$ were compared with the ground-truth dimensions (x, y, z) obtained by manual measurement, where all values are expressed in meters (m). The normalized size error was defined as

$$E_{\text{norm}} = \frac{(|\hat{x}-x|+|\hat{y}-y|+|\hat{z}-z|)/3}{\max(x,y,z)} \times 100. \quad (1)$$

Fig. 5 shows the overall error distribution. **Fig. 5(a)** presents box plots for each category. **Fig. 5(b)** shows a histogram of all instances. The mean normalized size error was 11.96%, indicating that the proposed pipeline can provide an initial scale estimate that is coarse yet usable for many everyday objects. Most categories remained within a moderate error range. However, the bottle category exhibited larger errors and variance compared to the others. Further inspection revealed that during language-assisted object analysis, a plastic water bottle was misidentified as a spray bottle. This led to an incorrect shape prior and an overestimated size prediction. These results suggest that, although semantic assistance is helpful for rough scale estimation, the process remains sensitive to category confusion among visually similar objects.

In terms of efficiency, generating 32 models simultaneously on an RTX 5090 yielded an overall throughput of about 2.59 objects per hour. The texturing stage accounted for most of the processing time. These results suggest that texture Generation, rather than geometric initialization, is the main issue with the current implementation.

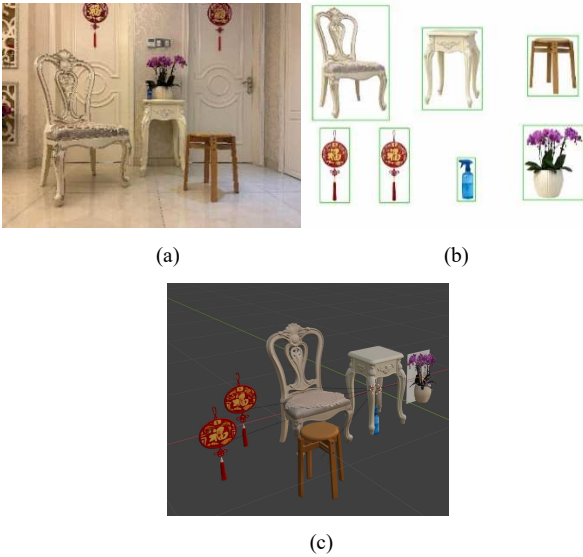


Fig. 4. Qualitative example of asset Generation from a real household scene: (a) input image, (b) extracted movable objects, and (c) generated 3D assets arranged for simulation initialization.

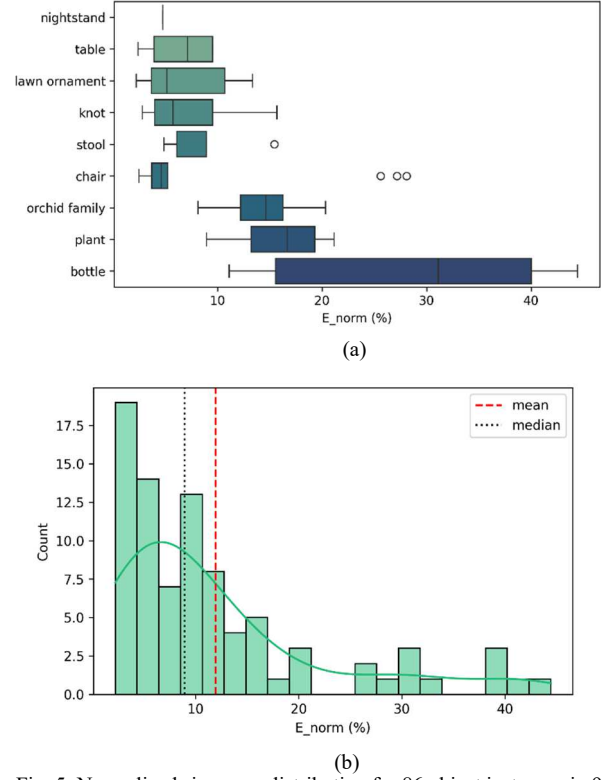


Fig. 5. Normalized size error distribution for 86 object instances in 9 categories: (a) category-wise boxplots of E_{norm} , and (b) overall histogram of E_{norm} , with the mean and median indicated.

B. Evaluation of Basic Object Placement

To evaluate the system's basic placement capability with natural-language input, we entered the instruction "place a cup on a table" and repeated the experiment 50 times in the same environment. The system automatically generated a scene-layout JSON file for each trial and then produced the corresponding 3D arrangement. We compared the generated scenes with a reference scene in terms of absolute position, relation consistency, and physical plausibility. Since this experiment contains only a single support relation, it is intended to validate text-guided object placement basics rather than evaluate complex scene-layout reasoning comprehensively.

The quantitative results are summarized in **Fig. 6**. **Fig. 6(a)** shows that the mean object position error was 9.357 mm (0.009357 m), and the cup position error was 1.8715 cm (0.018715 m). The supporting table remained fixed across all trials, with a center-position error of 0 mm, indicating that the variation mainly came from the placement of the cup. The mean pair center-distance error was 0.698 mm (0.000698 m), suggesting that the relative spatial structure was preserved with good repeatability, though small in-plane deviations remained in fine-grained alignment.

As shown in **Fig. 6(b)**, the spatial relation accuracy reached 100%, meaning that the contact relation between the cup and the table was preserved in all 50 trials. The collision-free rate was also 100%, indicating that no obvious geometric conflicts occurred in this simple single-support scenario. Overall, for a basic natural-language placement task such as the table-cup example, the system can generate semantically correct and collision-free initial scenes with good repeatability. However, this experiment does not yet

demonstrate robustness for dense clutter, multiple support chains, or more complex relational instructions.

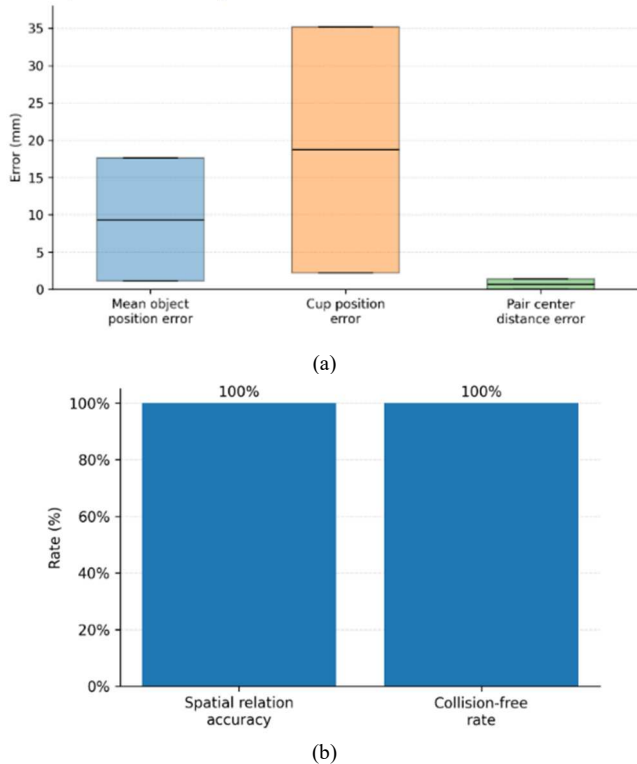


Fig. 6. Quantitative results of the natural-language-guided placement experiment over 50 trials: (a) distribution of positional and pairwise center-distance errors, and (b) spatial relation accuracy and collision-free rate.

IV. CONCLUSION

This paper presents a practical Real-to-sim pipeline for generating simulation-ready initial scenes from a single photograph. The proposed method focuses on movable, interactive objects and integrates the following processes: object extraction; 3D asset Generation; initialization of coarse semantic and physical attributes; orientation correction; and automatic layout Generation. The experimental results demonstrated the framework's preliminary feasibility. In the asset-Generation experiment, the system provided usable, object-scale priors for initializing the simulation.

In the placement experiment, it maintained a simple support relationship with good repeatability and no collisions in a controlled scenario. While the current throughput is limited by the texture-Generation stage, the system's ability to automatically assign physical attributes significantly reduces the specialized manual labor required for environment modeling. These results suggest that the proposed pipeline can reduce a substantial part of the manual effort required to prepare simulation-ready assets and layouts from real observations. This bridge between raw visual data and physics-based simulation provides a scalable foundation for large-scale robotic training. Future work will extend the framework to more complex multi-object scenes, improve geometric and physical accuracy, and strengthen its use in digital-twin-based robot simulation.

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